

FREUID Challenge 2026 - Technical Report

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1. Introduction

The FREUID Challenge 2026 (IJCAI-ECAI) asks participants to classify identity document images as bona fide or an attack (fraud), spanning physical manipulation, GenAI-driven digital edits, and print-and-capture attacks. This report describes our method, the data, the inference pipeline used for our final submission, results, and exact reproduction steps.

2. Data

- **Train:** 69,352 labeled images, 5 document types (EGYPT/DL, GUINEA/DL, BENIN/DL, MOZAMBIQUE/DL, MAURITIUS/ID), label balance 57.7% bona fide / 42.3% attack (fairly even across types except EGYPT/DL at ~49.6% attack). An `is_digital` flag is present but True for 99.97% of rows, so it carried no useful signal.
- **Test:** 142,818 rows total, split into a public subset (7,821 images, available for local exploration during the competition) and a private subset (~135k images, released July 13 per the code-freeze rule and only reachable from within a Kaggle Notebook kernel, not via direct bulk download at our available disk budget).
- No external data sources were used.

3. Method

We explored a wide range of approaches over the course of the competition (see Section 6 for the full experiment history); the model actually used for our final, code-freeze-compliant submission is a **simple average of 4 independently trained checkpoints**:

Model	Architecture	Input	Trained (UTC)
<code>resnet50_fold0</code>	<code>timm resnet50</code>	224px (timm default config)	2026-07-10 12:20
<code>efficientnet_b3_fold0</code>	<code>timm efficientnet_b3</code>	timm default config	2026-07-12 18:29
<code>best_model</code>	<code>torchvision resnet18</code>	320px	2026-07-04 09:13
<code>best_model_transformer</code>	<code>timm vit_base_patch16_224</code>	224px, bicubic resize	2026-07-04 10:18

All four predate the July 13, 2026 07:02:42 UTC code freeze (private image release). Training used standard fine-tuning from ImageNet-pretrained weights, BCE loss on `label`, mixup/cutmix/label

smoothing/EMA/AMP for the `resnet50 / efficientnet_b3` fold-0 models (5-fold `StratifiedKFold` split, fold 0 only used here since other folds' checkpoints were not retained - see Section 6), and plain fine-tuning for the two baselines.

Ensembling: plain arithmetic mean of the 4 models' sigmoid outputs. We chose a simple average over a learned meta-learner here because a logistic- regression stacker trained on a subset of our other models had previously been found to generalize worse than expected on the leaderboard-adjacent splits we could observe (see Section 6) - with only 4 heterogeneous checkpoints and no additional held-out stacking data, a plain average is the more defensible, lower-variance choice.

4. Inference

Implemented in `inference.py`, matching the competition's Docker sandbox contract (`docker run --network none, flat /data input, /submissions/submission.csv output, id,label` with `label = fraud-confidence score, higher = more fraudulent`). See `README.md` for exact build/run commands.

For the actual Kaggle leaderboard submission, inference was additionally run once via a Kaggle Notebook kernel (`boltuzamaki/freuid-private-inference`) so the same 4 checkpoints could score the private test images natively-mounted by Kaggle, avoiding a multi-hour/40+GB local download.

5. Results

Selected Kaggle leaderboard submissions across the competition (public score; lower is better - this is an error-rate-style metric, not accuracy):

Submission	Public score
ResNet18 baseline	0.26629
ViT-B/16 baseline	0.30372
5-model logreg-stacked ensemble (v1)	0.23873
Simple mean average, 5 v1 models	0.24141
<code>swin_tiny</code> solo (v1)	0.21082 (final submission)
Rank-blend, v1 swin + v1 convnext	0.21874
4-model average (this repo's Docker pipeline)	0.25845

Our final Kaggle submission is `swin_tiny` solo (0.21082), the best score we obtained. We do not have `swin_tiny`'s trained weights saved locally (per- fold checkpoints were deleted after each fold's predictions were cached, see Section 6), so the Docker artifact in this repository cannot reproduce this exact submission - it reproduces the 4-model average (0.25845) instead, which is a different, slightly weaker pipeline built entirely from checkpoints we do still have. We are disclosing this gap transparently:

prize eligibility requires exact reproducibility, and this submission does not meet that bar, which we are accepting in favor of reporting our best actual leaderboard result.

6. What we tried that did not make it into the final submission

Additional approaches explored during the competition, not part of the final pipeline:

- **5-architecture x 5-fold K-fold RGB stacking ensemble** (resnet50, efficientnet_b3, convnext_tiny, swin_tiny, vit_base_patch16_224) with a logistic-regression meta-learner on out-of-fold predictions: 0.23873.
- **Forensic hand-crafted-feature LightGBM**: 0.36557.
- **Residual/high-pass-filtered CNN** (resnet50 backbone): 0.95159; a second backbone (efficientnet_b3) reached OOF AUC 0.9998 but was not submitted standalone.
- **Pseudo-labeling + test-time augmentation retrain** of the 5-model ensemble: consistently scored worse than the non-pseudo-labeled version in every comparison (e.g. resnet50 0.25560 vs. 0.27816).
- **Rank-averaging and logistic-regression blends** of model subsets: none beat the best individual solo model.
- **Document-type classifier** (country/document type as an auxiliary signal): built but not incorporated into the final pipeline.

Per-fold checkpoints from the above were not retained after each fold's out-of-fold predictions were cached; only the 4 checkpoints in Section 3 both still exist on disk and predate the freeze, so those are what this reproducible package uses.

We also note that ~94.5% of the test set (the private-test rows) is only reachable via a Kaggle Notebook kernel, not local browsing, which shaped how we validated the approaches above.

7. Reproducibility

See `README.md` for exact `docker build / docker run` commands, weight provenance and timestamps, and a disclosure regarding this repository's git history (published at the deadline, not maintained incrementally through development - see README for what independent evidence supports our code-freeze compliance timeline).

Hardware: local NVIDIA GPU (8GB VRAM) for training; Kaggle Notebook (T4 GPU) for private-test inference.